

Harun Emrecan Karabağ

Unity / Unreal Game Developer | C# Software Engineer | Gameplay & Systems

Ankara, Türkiye | +90 507 049 3574 | harunemrecankarabag@gmail.com

Portfolio: hek75.com | Itch.io: hekyetmisbes.itch.io | GitHub: github.com/hekyetmisbes | LinkedIn: linkedin.com/in/harunemrecankarabag

Professional Summary

Computer Engineering graduate and current M.Sc. student in Computer Animation and Game Technologies, focused on Unity/C#, Unreal Engine 5 gameplay prototyping, gameplay systems, software architecture, mobile development, UI/HUD implementation, simulation, and interactive product development. Hands-on experience spans psychological and narrative puzzle design, casual and mobile games, driving simulation, backend software engineering, research-oriented simulation, data persistence, third-party services, testing, and performance optimization.

I have developed projects from early concept and technical architecture through implementation, debugging, validation, iteration, and published playable builds. My engineering approach combines clean and reusable code, data-driven design, algorithmic problem solving, player-facing polish, technical documentation, and performance-conscious development. I am comfortable taking ownership in small multidisciplinary teams while continuing to deepen my skills in scalable game systems and software engineering.

Professional Focus

- End-to-end gameplay development: Unity/C# production, Unreal Engine 5/Blueprint prototyping, progression, persistence, UI/HUD, input, testing, optimization, and build preparation.
- Systems-oriented problem solving: data-driven architecture, state management, event-driven flows, algorithms, AI/pathfinding, validation, reusable interaction systems, and performance-aware runtime design.
- Product and player experience: rapid prototyping, readable interaction loops, responsive feedback, mobile-first interaction thinking, puzzle/narrative pacing, game jam production, and iterative polishing.
- Software engineering discipline: OOP, SOLID fundamentals, modular code, Git workflows, backend development, documentation, debugging, testing, and maintainable project organization.

Core Skills

Game Development: Unity 6, Unity 2D/3D, Unreal Engine 5.5, C#, C++ fundamentals, mobile and casual games, simulation prototypes, narrative/puzzle games, platformers, game jam prototyping

Gameplay & Systems: player movement, interaction states, room/mission flow, scoring, progression/economy foundations, achievements, skill trees, daily challenges, collectibles, win/loss logic, game-state flow, touch interaction language

Unity UI/UX Integration: Unity UI, Canvas, menus, HUD elements, score/result/settings screens, responsive UI thinking, Figma-to-Unity workflow, player feedback systems

Unreal Engine: Blueprint gameplay logic, interaction/state prototyping, level/blockout workflow, environmental puzzle design, player feedback, C++ learning foundations

Architecture & Code Quality: OOP, SOLID fundamentals, modular architecture, composition root, ScriptableObjects, state machines, event-driven systems, interfaces, serialization/migration, Repository Pattern, Dependency Injection, CQRS fundamentals

Algorithms & AI: A* pathfinding, graph traversal, heuristic cost design, spatial-grid indexing, Dictionary, HashSet, Queue, Stack, priority queues, agent-based simulation, optimization, forecasting, AI-supported modeling

Testing & Debugging: Unity Test Framework, EditMode/PlayMode test foundations, xUnit, validation, bug reproduction, debugging, refactoring, regression-testing concepts, QA checklist mindset

Performance: object pooling, spatial partitioning, distance-based update throttling, performance-aware update loops, allocation-aware architecture, profiling concepts, CPU/GPU fundamentals

Services, Data & Backend: Firebase Realtime Database, UnityWebRequest, SQLite, JSON persistence, ASP.NET Core 8, Entity Framework Core, FluentValidation, data modeling, logging

Tools & Workflow: Git, GitHub, GitLab, Visual Studio, Docker fundamentals, Figma, technical documentation, build testing, iterative development

Languages: Turkish - Native; English - B2+ written and spoken

Selected Game & Software Projects

The Final Loop - Unreal Engine 5 Psychological Puzzle / Vertical Slice

[Project Details](#)

Single-player psychological escape-room experience set inside an abandoned research facility, where room-based puzzle progression gradually exposes a suppressed memory loop, identity conflict, and the illusion of escape. The current production target is a focused 20–30 minute PC vertical slice.

Technologies: Unreal Engine 5.5.4, Blueprints, interaction-state gameplay, environmental puzzles, lightweight inventory, narrative systems, UI/feedback, Git

- Developing a 3–4 room vertical-slice flow that ties spatial/logic puzzle solutions directly to narrative revelation instead of treating puzzles as isolated gates.
- Building Blueprint-driven interaction patterns for environmental inspection, clue collection, doors/terminals, notes and recordings, trigger events, gameplay states, and readable player feedback.
- Designing a lightweight inventory/object-combination layer and a reinforcement-pill / flashback mechanic that connects progression with memory fragmentation and changes how the player interprets the facility.
- Structuring project content under dedicated Core, Interaction, Puzzles, Maps, Audio, UI, Narrative, and Data areas to keep collaboration and feature growth maintainable.
- Using repeated attempts, failure/retry rhythm, level flow iteration, debugging, and blockout experimentation as a project-based Unreal Engine learning and production workflow.

The Unlit Door - Mobile-First Psychological Puzzle / Adventure

[Project Details](#)

Chapter-based psychological narrative game concept that converts childhood fear, shame, silence, and accumulated emotional pressure into a consistent mobile interaction language. The player advances through four memory doors, with each chapter reinterpreting the same inputs in a different emotional context.

Technologies: Unity 6 (6000.3.12f1), C#, URP, Unity Input System, mobile interaction design, narrative/puzzle design, atmosphere and feedback

- Defining a shared tap / hold / swipe control language for reaction timing, panic-state inputs, escape/repositioning, breathing control, and emotionally readable mobile interactions.
- Designing the planned chapter flow across The Chase, The Match, The Stage, and The Unlit Door, with each section using mechanics and pacing to express a distinct psychological pressure.
- Preparing the vertical-slice structure around a memory-center hub, chapter grayboxes, atmosphere, audio, touch feedback, and a scalable project organization under a dedicated game-content root.

Honey Tycoon - Unity 6 Casual / Idle Systems Foundation

[Portfolio](#)

PC/mobile-targeted idle management project built around data-driven production, upgrades, research, automation, economy, and long-term progression systems.

Technologies: Unity 6, C#, URP, ScriptableObjects, Unity Test Framework, JSON, Git

- Designed a scalable technical foundation with versioned GameState, Profile, Economy, Inventory, and World models to keep progression data explicit and maintainable.
- Implemented JSON serialization and migration foundations, a bootstrap composition root, ScriptableObject-based definitions, and editor-side validators for configuration and state integrity.
- Separated runtime, editor, and test assemblies and documented architectural/performance rules for atomic economy transitions, frame-rate-independent production, save/version compatibility, and mobile performance budgets.

Delivery Driver - Open-City Delivery Simulation Prototype

[Project Details](#) | [GitHub](#) | [Playable](#)[Build](#)

Stylized Unity 3D driving and delivery project combining player control, mission/progression systems, route guidance, minimap navigation, NPC traffic, persistence, UI/HUD feedback, and runtime optimization.

Technologies: Unity 3D, C#, Input System, Cinemachine, A*, spatial indexing, SQLite, EasyRoads3D, NPC traffic AI, object pooling, Git

- Built delivery missions, pickup/drop-off flow, money and XP rewards, levels, achievements, skill-tree progression, daily challenges, cargo damage, statistics, settings, and save/load flows.
- Implemented incremental A* road-graph pathfinding using Dictionary/HashSet search state, queued priority handling, heuristic costs, waypoint graphs, and spatial-grid indexing.
- Developed route/minimap guidance, HUD feedback, NPC traffic, obstacle avoidance, pooled spawning, spatial partitioning, distance-based update throttling, and simplified far-AI behavior.
- Published a downloadable Windows build and maintained the project through iterative updates, bug fixes, gameplay improvements, and development logs.

Selected Game & Software Projects (continued)

Fog Bridge - Atmospheric Narrative Puzzle Game

[Project Details](#) | [Playable Build](#)

GameDev.tv Game Jam 2026 project about two isolated guards gradually building understanding through paper-plane messages across a fog-covered chasm.

Technologies: Unity, C#, narrative puzzle design, fictional alphabet decoding, interaction flow, resource gathering, bridge rebuilding, Itch.io release

- Built a compact interaction loop where the player receives paper-plane messages, decodes a fictional alphabet, replies across the fog, gathers materials, and gradually rebuilds a forgotten bridge.
- Focused on emotional pacing, symbolic worldbuilding, readable puzzle feedback, resource/progression triggers, and interaction-driven narrative advancement.
- Completed and published the project within a game-jam production cycle, demonstrating rapid scope control, testing, polishing, and release discipline.

Flag Quiz - Android Mobile Quiz Game

[Project Details](#) | [GitHub](#)

Mobile quiz game where players identify country flags through a simple, accessible, score-driven loop with remote data integration and mobile-focused UI.

Technologies: Unity 2D, C#, Android, Firebase Realtime Database, UnityWebRequest, Unity UI, Figma, Git

- Built randomized flag presentation, answer validation, score tracking, round progression, success/failure states, audio feedback, and high-score persistence.
- Integrated Firebase Realtime Database for remote question data and UnityWebRequest for flag images and remote content retrieval.
- Designed and iterated Android-focused UI interactions around clarity, responsive input, short-session usability, and readable player feedback.

The Platform - 2D Platformer Game

[Project Details](#) | [GitHub](#) | [Playable Build](#)

Released cyberpunk-themed 2D platformer built around movement, hazards, collectibles, level progression, and straightforward completion goals.

Technologies: Unity 2D, C#, platformer mechanics, level flow, collectibles, ShaderLab/HLSL, Git

- Developed 2D movement, obstacle interaction, level progression, collectible-driven objectives, and completion conditions for a complete playable loop.
- Designed levels around collecting stars, navigating hazards, and completing objectives with clear visual/gameplay feedback.
- Explored ShaderLab/HLSL alongside core Unity development and published a downloadable Windows build with install instructions.

Movidle Game - Movie Guessing Game

[Project Details](#) | [GitHub](#)

Data-backed guessing game based on IMDb Top 250 information, comparing player guesses through attributes such as genre, year, director, country, and star.

Technologies: Java, game logic, validation systems, CSV/data handling, UI flow

- Developed the core guessing logic where the system randomly selects a target movie and compares player predictions across multiple movie attributes.
- Implemented limited-attempt gameplay, feedback-driven comparisons, success/failure states, and CSV-backed movie selection.
- Used the project as an early foundation for game-state logic, validation, data-driven behavior, and player feedback before later expanding into Unity/C# development.

Tiny Tide Harbor - Unity 6 Mobile Route Puzzle / Traffic Management

[GitHub](#) | [Portfolio](#)

Android-focused casual puzzle and route-management game built around one-finger route drawing, boat traffic, currents, docks, and collision-aware navigation in portrait orientation.

Technologies: Unity 6 (6000.3.12f1), C#, URP 17.3.0, New Input System 1.19.0, Android API 34+, ScriptableObjects, object pooling, EditMode/PlayMode tests

- Designed the gameplay architecture around route drawing, boat movement, collision detection, current behavior, dock logic, and traffic-management decisions suited to short mobile sessions.
- Organized the project into modular Core, Data, Gameplay, UI, Audio, Monetization, Analytics, and test areas, using ScriptableObject-backed boat/dock/current/level data plus pooling and signal foundations.
- Targets a portrait Android build at 60 FPS with a zero-GC gameplay-loop goal, supported by automated EditMode/PlayMode test areas for gameplay math and logic validation.

Professional Experience

Co-Founder Fiktion Technology Studios Ankara, Türkiye	Dec 2025 - Apr 2026
Contributed to early-stage game and software initiatives through technical direction, feature planning, prioritization, documentation, implementation ownership, and coordination in a small multidisciplinary team.	
Software Engineering Intern Erciyes University Research Dean's Office / IT Kayseri, Türkiye	Jul 2025 - Aug 2025
- Developed maintainable ASP.NET Core 8 and Entity Framework Core components with data modeling, validation, debugging, logging, documentation, and iterative improvements. - Applied Clean Architecture, CQRS, Repository Pattern, Dependency Injection, FluentValidation, Docker fundamentals, xUnit tests, and Git-based collaborative development practices.	
Hardware Engineering Intern Fotonik Teknoloji A.Ş. Kayseri, Türkiye	Jul 2024 - Aug 2024
Supported hardware system design and improvement, hands-on testing, system analysis, issue investigation, and technical problem solving in a cross-disciplinary engineering environment.	
Research Intern - 2247-C STAR Fellow TÜBİTAK Türkiye Advisor: Asst. Prof. Saltuk Buğra Selçuklu	Feb 2024 - Jul 2024
Worked on agent-based simulation, optimization, forecasting, and AI-supported modeling, strengthening algorithmic thinking, experimentation, quantitative reasoning, and structured problem solving.	

Education

Hacettepe University M.Sc. in Computer Animation and Game Technologies GPA: 3.81 / 4.00 Expected Graduation: Jan 2027	2026 - Present
Erciyes University B.Sc. in Computer Engineering GPA: 3.00 / 4.00	2020 - 2025

Certificates

- Introduction to Game Design - Epic Games / Coursera - Nov 2025
- Unity 401 - Turkcell Geleceği Yazanlar - Jul 2024
- More C# Programming and Unity - University of Colorado Boulder / Coursera - Mar 2024
- C# Class Development - University of Colorado Boulder / Coursera - Mar 2024
- Introduction to C# Programming and Unity - Coursera
- Version Control: Git and GitHub - BTK Akademi - Oct 2023
- Hypercasual Fundamentals - BTK Akademi - Jan 2023